

STEP Mock Test

Date: 14/09/2026 | Time: 45 Minutes | Max Marks: 45

Section A: Aptitude and Numerical Reasoning - $2 \times 5 = 10$ Marks

1. [5 Marks]

Convert **3475** from decimal to octal.

2. [5 Marks]

A shopkeeper buys **150 notebooks at ₹24 each**. He sells **60 notebooks at ₹30 each** and the remaining notebooks at **₹20 each**. Determine whether he makes a **profit or loss**, and calculate the amount.

Section B: C Programming - $5 \times 5 = 25$ Marks

3. [5 Marks]

Write a C program to check whether a given character is present in the following character array. The comparison must be **case-sensitive**.

```
char arr[] = "Programming";  
char ch = 'g';
```

Print **Yes** if present, otherwise print **No**.

4. [5 Marks]

Write a C program using the following two arrays to print the elements of the **first array that occur exactly three times** in the second array.

```
int first[] = {2, 4, 6, 8, 10, 12};  
int second[] = {2, 4, 6, 6, 8, 10, 6, 12, 10, 10};
```

Expected output:

```
6 10
```

5. [5 Marks]

Write a C program to print all numbers between **40 and 100** that are:

- divisible by **5**
- not divisible by **10**

6. [5 Marks]

Write a C program to print the following series:

```
2 4 6 8 16 10 32
```

Identify the pattern and implement it using a loop.

7. [5 Marks]

Write a C program to print **Yes** if **any element in the array is exactly double its next element**.

```
int arr[] = {3, 12, 6, 24, 12, 7};
```

For the given array, the output should be:

```
Yes
```

Section C: Code Tracing - $2 \times 5 = 10$ Marks

8. [5 Marks]

Trace the following C program and write the **exact output**. Show the important intermediate values.

```
#include <stdio.h>

int check(int n)
{
    return n % 5;
}

int main(void)
{
    int arr[] = {14, 21, 18, 25, 32};

    for (int i = 0; i < 5; i++)
    {
        if (check(arr[i]) == 1 || check(arr[i]) == 2)
        {
            printf("%d ", arr[i] + 2);
        }
    }
}
```

```
    return 0;
}
```

9. [5 Marks]

Trace the following program **step by step** and write the exact output.

```
#include <stdio.h>

int main(void)
{
    int total = 0;

    for (int i = 1; i <= 2; i++)
    {
        for (int j = 0; j <= 2; j++)
        {
            printf("%d ", i + j);

            if (i + j <= 3)
            {
                total += i;
                printf("%d ", total);
            }
        }
    }

    return 0;
}
```